SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

# ASSESSMENT TOOL 1 – Concept Mapping Questions

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| Course Code: | CSA0315 | Course Title: | Data Structures |
| CO Assessed: | CO1 | Assessment: | Assessment Tool 1 – Concept Mapping Questions |
| Weightage: | 10% of CIA | Total Marks: | 100 |
| CO1 Statement: | Basics, Data, Data object, Data structure, Abstract Data Types (ADT), realization of ADT in 'C', Primitive and non-primitive data structure, Linear and non-linear data structure Arrays & Recursion, Performance analysis and Measurement: Time and space analysis of algorithms, Average, best and worst-case analysis. | | |
| Student Name: | K.B.Tharunika | Reg. No.: | 192525276 |
| Section / Batch: | Slot D | Date: | |

## Code of Conduct

*I K.B. Tharunika(192525276)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student:K.B.Tharunika

## Instructions

- This test contains 3 Concept mapping questions. Total: 100 Marks.
- When a student answer using a concept map, in evaluation the answer should contain following four requirements: Understanding of concepts, Connections between ideas, Logical structure, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.
- No DTP printouts or electronic devices permitted. They should provide written materials.

| Section / Topic | Focus | Q Nos | Marks |
|---|---|---|---|
| Basics, Data, Data object, Data structure, Abstract Data Types (ADT), realization of ADT in 'C' | Concept mapping of Basics in Data Structure | 1 | 30 |
| Performance analysis and Measurement: Time and space analysis of algorithms, Average, best and worst-case analysis. | Fibonacci series | 2 | 40 |
| Primitive and non-primitive data structure, Linear and non-linear data structure Arrays | Concept mapping on classifications of data structures. | 3 | 40 |
| | | GRAND TOTAL: | 100 Marks |

Name : K.Tharunika
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Course Name & code: CSA-0315 Datastructures

# Assignment-1

Co-1 AT-1

1. write a C program to count no of odd & even elements in an array of 'n' elements

Aim:-

To write a C programme count the num of odd & even elements(n)

18/20

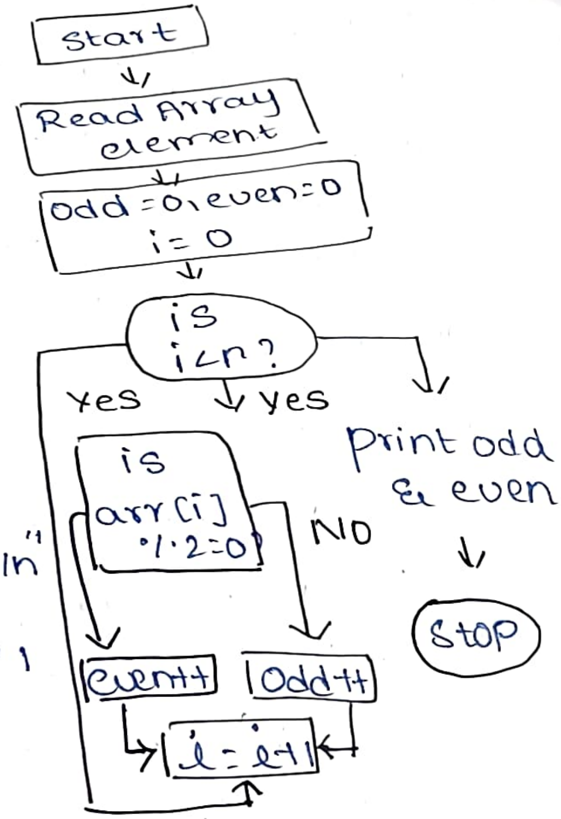
Algorithm:-

1. Start
2. Read the no of elements(n)
3. Read the array elements
4. Intilaize odd=0 & even=0
5. Check each element
6. Display the no of odd & even element

Program:-

```
#include <stdio.h>
int main()
{ int n, i, a[100];
int add=0, even=0;
printf("enter no of
        elements:");
scanf("%d", &n);
printf("enter array elements:\n");
```

```
for (i=0; i<n; i++)
{ scanf ("%d", &a[i]);
}
for (i=0; i<n; i++)
{ IF (a[i] % 2==0)
  even++ ;
 else
   odd++ ;
}
printf("even elements =%d\n"
          , even);
printf ("odd elements=%d\n"
          , odd);
return 0;
}
```




Output:-

Enter no of elements = 5

Enter no. of array element : 10 15 8 7 20

Even = 3

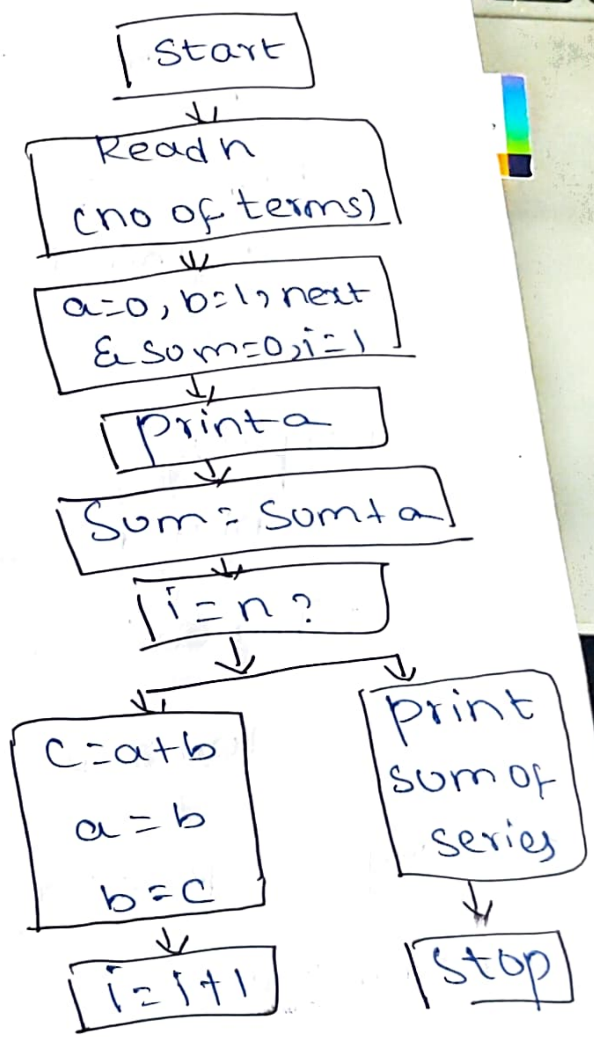
odd = 2

2. c programme to print fibanocci series & find its sum.

Aim:- To write a c programme to print the fibanocci series & calculate its sum

Algorithm:-

1. Start
2. Read the no of terms
3. Intilize a=0, b=1, sum=0
4. print Fibanocci term
5. Add Each term to sum
6. Display the total sum
7. Stop




Program:-

```
# Include <stdio.h>
    int main( )
  { int n, i;
   int a=0, b=1, c;
     int sum=0;
printf("enter no of terms:");
Scanf("%d", &n);
printf("fibanocci series:/n");
```

```
for(i=1; i<=n; i++)
{ printf("%d", a);
printf("%d", a)
sum = sum + a ;
    c = a+b;
    a = b;
    b = c;
}
printf("\nsum=%d", sum);
return 0;
}
```

Output:-

Enter no of element : 7

Fibanocci series: 0 1 1 2 3 5 8

Sum = 20

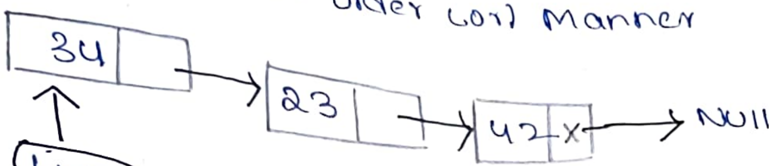
3. Concept map for data structures & classification

Definition:-

A data structure is a way of organisation storing & managing data so that it can be accessed & modified efficiently.

## Linked List:-

First in First Order (or) Manner




Head singled linked list

| data | Address |
|---|---|

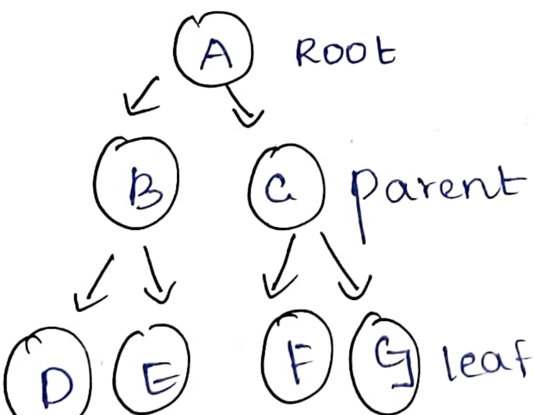
Node

## NON linear datastrcuctore:-

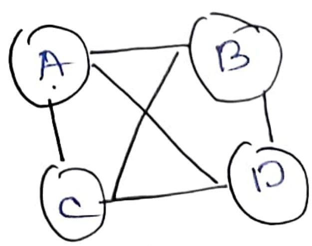
↳ NON sequential order

1. Tree
2. Graph

## Tree:-




## Graph:-




G = {V, E}

V = {A, B, C, D}

E = {AB, BD, AC, CD, AD}

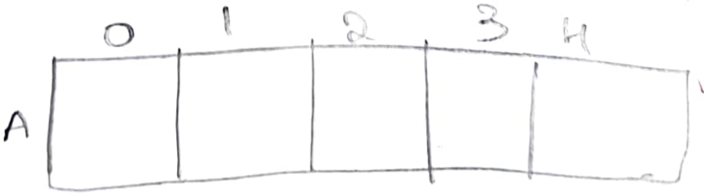
# Classification of data structure

- Premittive data Structure
- Non premitive datastructure
  - Linear
    - Static
      - Array
    - Dynamic
      - Stack
      - Queue
  - Non linear
    - Tree
    - Graph

* int, float, string.char
* pointer is a variable
* Structure (collect at datatype)
* It contains, basic data strudures

**Array:-** collection of location in memory under the same name & data type is called array

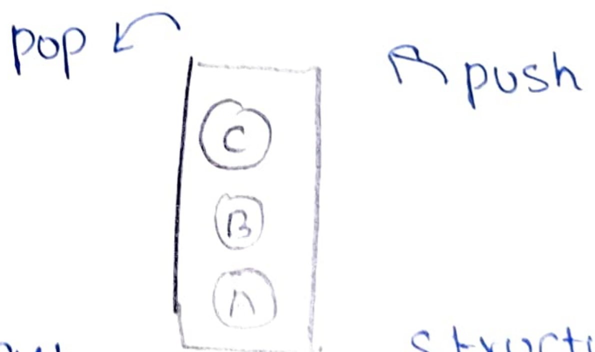
Ex:-




Array in data Structure

**Stack:-** Last in first manner (The boundary of data structure is fixed.

**Nonstack:-** No boundary & memorylocation are not fixed




Stack of Data Structure (LIFO)

## Annexure A – Application Based Assessment

1) Write a c program and draw the flow chart as concept map for finding the count of odd and even numbers in an array of n numbers.

2) Write a concept map based flow chart and also write the program to generate the Fibonacci series and sum of the series up to n numbers.

3)A smart city traffic management system handles data related to vehicle movement, road networks, and signal control. Construct a concept map connecting Primitive Data Structures → Non-Primitive Data Structures → Linear → Non-Linear Structures. Include examples such as arrays, linked lists, trees, and graphs, and illustrate their applications in traffic management. Explain the relationships between these structures and justify their classification based on functionality and efficiency.

### Rubrics for Calculation

| Criteria | Weightage | Level 4 – Exemplary | Level 3 – Proficient | Level 2 – Developing | Level 1 – Beginning |
|---|---|---|---|---|---|
| Understanding of Problem Context | 20 | Clearly understands the problem and accurately identifies all requirements | Good understanding with minor gaps | Basic understanding; some requirements unclear | Poor understanding or misinterpretation of the problem |
| Application of Concepts | 30 | Accurately applies appropriate concepts to solve complex problems | Applies relevant concepts correctly with minor errors | Applies basic concepts with limited accuracy | Fails to apply appropriate concepts |
| Problem-Solving Approach | 20 | Logical, well-structured, and efficient approach | Mostly logical approach with minor gaps | Basic approach with some inconsistencies | Illogical or unclear approach |
| Accuracy of Solution | 20 | Completely correct solution with proper justification | Mostly correct with minor errors | Partially correct solution | Incorrect or incomplete solution |
| Clarity | 10 | Clear, well-organized, and properly explained solution | Understandable with minor clarity issues | Limited clarity and explanation | Poorly presented and difficult to understand |

| Faculty In-charge | Course Coordinator | Program Director |
|---|---|---|
| | Signature: | Signature: |